DeckScan is a crowdsourced, real-time parking heatmap for Monash students that eliminates the frustration of searching for a spot by showing live availability before you arrive.

Project Summary:

The Problem: For hundreds of students at Monash University, the daily commute is marred by the stress of finding a parking spot at the Sports Precinct. This leads to late arrivals for lectures and increased campus congestion. Currently, there is no way for a student to know if the SE4 multi-level deck is full until they are already circling it.

The Solution: We built DeckScan, a "Code for Community" project that provides a live visual heatmap of parking occupancy. By allowing users to check in and out of zones, the app crowdsources data to update a global map instantly. This allows students to check their phones before leaving home and decide whether to head to SE4 or the SE5 surface lot based on live availability.

How We Built It: The project was designed for simplicity and speed. The frontend uses standard HTML and JavaScript, utilizing the Leaflet.js library and OpenStreetMap for map rendering. We chose Firebase Realtime Database for the backend to ensure that any user action—like a check-in—reflects on every other user's screen in under one second. For the heatmap visualization, we implemented an IDW algorithm to translate raw occupancy numbers into a clear green-to-red visual gradient.

Challenges Faced: One of our primary hurdles was handling GPS accuracy and geographical rendering for a prototype. Because judges may be viewing the app from different locations, we had to program the parking zones to render relative to the user's current GPS coordinates rather than fixing them to Monash's actual coordinates. Additionally, we had to create a sophisticated simulation of "background drift" and historical trends to make the prototype feel alive and demonstrate how the app would look during peak hours, such as the 8–10 AM morning rush.

Links & Media

- **Public GitHub Repo:** github.com/professorrazan/deckscan
- **3-Minute Demo Video:** [Insert YouTube Link Here]